

RESEARCH STATEMENT

My research focuses on cost-friendly methods to make NLP models better assistants for people.

Research Interests: RAG, Memory, Personalization, Agent, NLP Applications

EDUCATION

Peking University

M.S. (Academic) in Computer Science, GPA: 3.60/4.00

Beijing, China

2024–2027

Peking University

B.S. in Computer Science, GPA: 3.72/4.00

Beijing, China

2020–2024

PUBLICATIONS

Refereed Publications

1. **Chenming Tang**, Hsiu-Yuan Huang, Weijie Liu, Clive Bai, Saiyong Yang, Yunfang Wu. “Do Not Step Into the Same River Twice: Learning to Reason from Trial and Error”. *ACL 2026 (Oral)*.
2. **Chenming Tang***, Yutong Yang*, Kexue Wang, Yunfang Wu. “Aligning Language Models with Real-time Knowledge Editing”. *ACL 2026*.
3. Hsiu-Yuan Huang*, **Chenming Tang***, Weijie Liu, Clive Bai, Saiyong Yang, Yunfang Wu. “Think Outside the Policy: In-Context Steered Policy Optimization”. *ACL 2026 (Findings)*.
4. **Chenming Tang***, Zhixiang Wang*, Hao Sun, Yunfang Wu. “Large Language Models Might Not Care What You Are Saying: Prompt Format Beats Descriptions”. *EMNLP 2025 (Findings)*.
5. **Chenming Tang**, Zhixiang Wang, Yunfang Wu. “SCOI: Syntax-augmented Coverage-based In-context Example Selection for Machine Translation”. *EMNLP 2024*.
6. **Chenming Tang**, Fanyi Qu, Yunfang Wu. “Ungrammatical-syntax-based In-context Example Selection for Grammatical Error Correction”. *NAACL 2024*.
7. **Chenming Tang**, Xiuyu Wu, Yunfang Wu. “Are Pre-trained Language Models Useful for Model Ensemble in Chinese Grammatical Error Correction?”. *ACL 2023 (Short Paper)*.

Selected Preprints

1. **Chenming Tang**, Jiawei Han. “Select-And-Extract: A Lightweight Plugin for Retrieval-Augmented Generation”. *arXiv:2608.00658* (under review).
2. **Chenming Tang**, Hsiu-Yuan Huang, Weijie Liu, Junqiang Zheng, Saiyong Yang, Yunfang Wu. “Democratizing Tool Learning with Environments Fully Simulated by a Free 8B Language Model”. *arXiv:2604.17739* (under review).
3. Kun Liang, **Chenming Tang**, Clive Bai, Weijie Liu, Saiyong Yang, Yunfang Wu. “ADWIN: Adaptive Windows for Horizon-Aware On-Policy Distillation”. *arXiv:2605.28396* (under review).
4. Kun Liang, Clive Bai, Xin Xu, **Chenming Tang**, Sanwoo Lee, Weijie Liu, Saiyong Yang, Yunfang Wu. “ORBIT: On-policy Exploration-Exploitation for Controllable Multi-Budget Reasoning”. *arXiv:2601.08310* (under review).
5. Hao Sun, **Chenming Tang**, Gengyang Li, Yunfang Wu. “Lost in the Passage: Passage-level In-context Learning Does Not Necessarily Need a ‘Passage’”. *arXiv:2502.10634*.
6. Fanyi Qu, **Chenming Tang**, Yunfang Wu. “Evaluating the Capability of Large-scale Language Models on Chinese Grammatical Error Correction Task”. *arXiv:2307.03972*.

* Equal contribution.

RESEARCH EXPERIENCE

University of Illinois Urbana-Champaign

Mar 2026 – Jul 2026

Advisor: Prof. Jiawei Han

Remote

- **Retrieval-augmented Generation:** Identified two fundamental failures of RAG: retrieval failure and reading failure; proposed a simple yet effective RAG plugin that directly addresses the two failures, outperforming advanced approaches at modest cost ([manuscript under review](#)).

Tencent & Peking University

Jul 2025 – May 2026

Advisor: Dr. Weijie Liu & Prof. Yunfang Wu

Beijing, China

- **Agentic Reinforcement Learning:** Showed that existing agentic RL approaches require labeled data, realistic tools, or environments synthesized or simulated by expensive proprietary LMs; democratized it by adaptive environments simulated by local open-source LMs, which is cost-friendly yet effective ([manuscript under review](#)).
- **Reinforcement Fine-tuning:** Revealed that current reasoning-oriented RL is limited by the upper bound of the LMs themselves, and existing mitigation strategies rely heavily on costly intermediate external guidance from human experts or advanced LMs; proposed a novel perspective by designing LTE, a paradigm that guides LMs with their own mistakes on problems beyond their initial capability, which boosts the reasoning performance of post-trained LMs while free from any expensive explicit external guidance (oral presentation at [ACL 2026](#)); also explored cheap in-context guidance from existing annotated data (published at [ACL 2026 Findings](#)).

Peking University

Mar 2022 – Jul 2025

Advisor: Prof. Yunfang Wu

Beijing, China

- **Knowledge Editing:** Identified that data resources are limited for real-time real-world knowledge editing, and existing methods are typically heavy or unreliable for this task; bridged the gap by developing CRAFT, a dynamic dataset for this task, and proposing KEDAS, a self-adaptive paradigm that need no parameter changes for editing and achieves significantly more reliable and balanced performance than prior approaches (published at [ACL 2026](#)).
- **Prompting and In-context Learning:** Pointed out current approaches for in-context demonstration selection mainly rely on lexical or semantic features and overlook syntax; introduced syntactic knowledge to demonstration selection for NLP tasks like grammatical error correction and machine translation (published at [NAACL 2024](#) and [EMNLP 2024](#)); also demonstrated that a suitable prompt format can achieve promising in-context learning performance on many tasks even without well-designed descriptions (published at [EMNLP 2025 Findings](#)).
- **Grammatical Error Correction:** Investigated how pre-trained LMs can help GEC and highlighted that LM perplexity does not necessarily correlate with grammatical correctness (published at [ACL 2023](#)).

AWARDS

- **Outstanding Graduate**, Peking University 2024
- **Exceptional Award for Academic Innovation**, Peking University 2023
- **Merit Student**, Peking University 2023
- **First prize**, NLPCC 2023 Shared Task 1 (Chinese Grammatical Error Correction) 2023
- **Second place**, CCL 2022 Shared Task 4 (Multi-reference Multi-source Chinese Learner Text Correction) 2022

TEACHING

Teaching Assistant | Introduction to Computing (C)

Spring 2025

Peking University | Instructor: Prof. Zhifang Sui & Prof. Yunfang Wu

Beijing, China

STANDARDIZED TESTS

- **TOEFL iBT:** 111 (R:30, L:28, S:23, W:30) Dec 2025
- **GRE General Test:** 336 (V:166, Q:170, AWA:3.5) Jul 2025